

Domain Adaptation for Sim-to-Obs Astrophysics: From DESI Mocks to J-PAS Observations



Daniel López-Cano



J-PAS seminar
12/2025

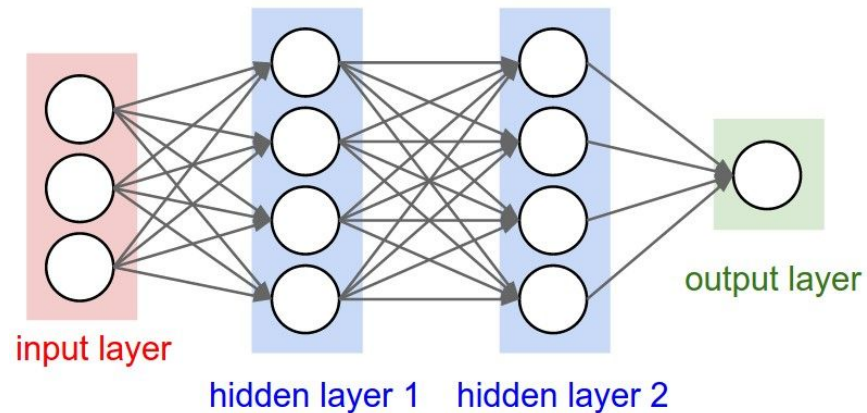
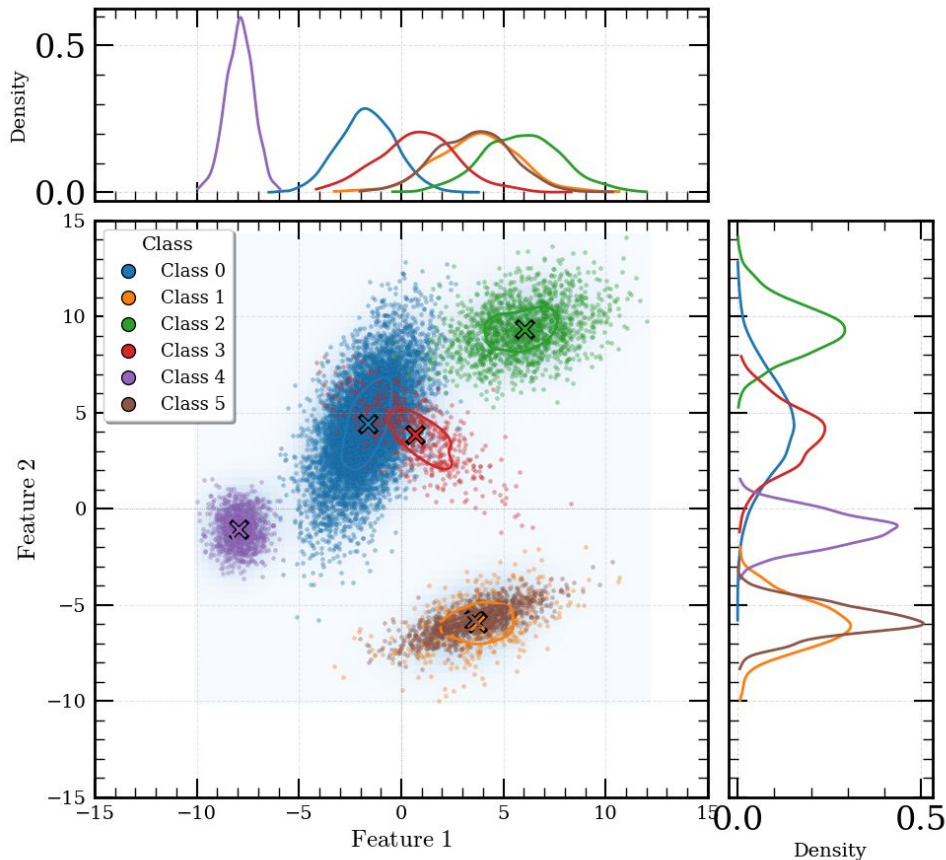


-
1. Problem: Domain Shift
 2. Solution: Adaptive/Contrastive techniques
 3. Application: From DESI mocks to J-PAS observations
 - 3.1 Data & Experiments
 - 3.2 Method
 - 3.3 Results
 4. Key Takeaways

1. Problem: Domain Shift

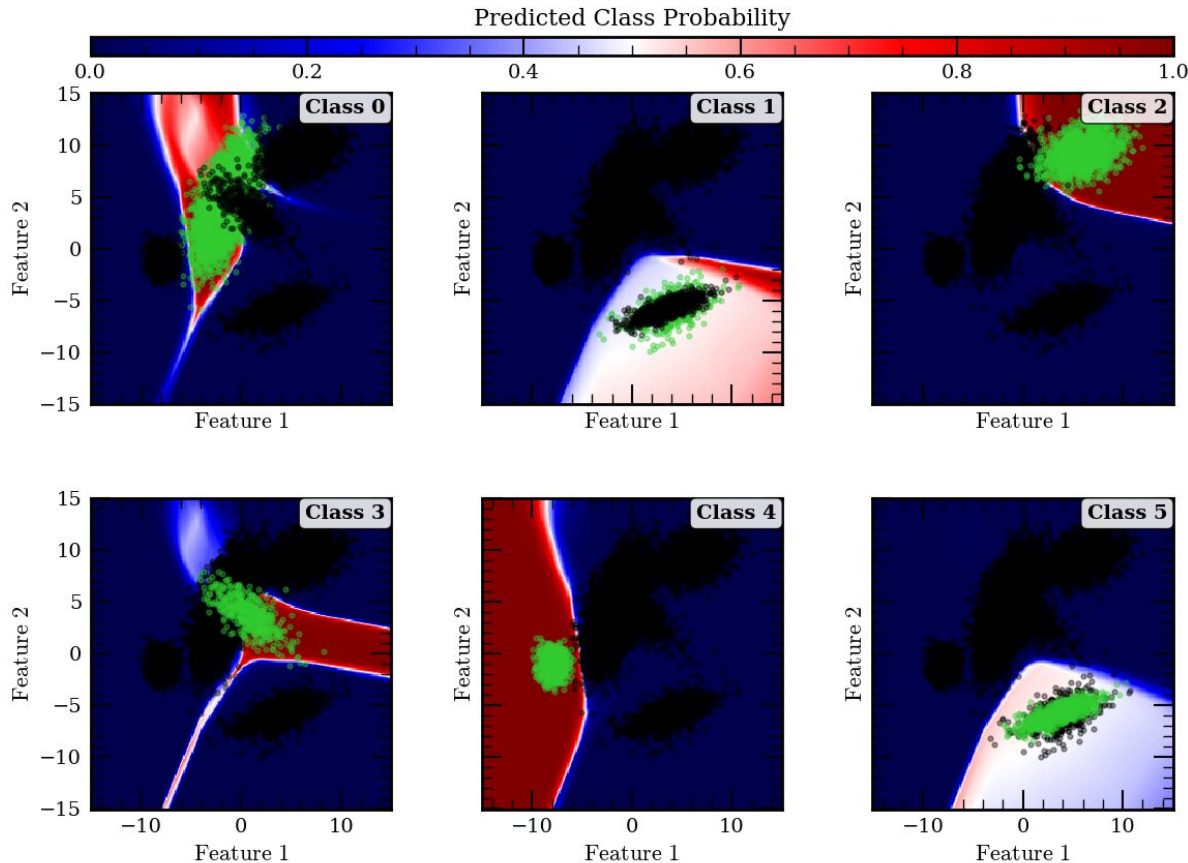
What happens when the Source and Target distributions differ?

1. Problem: Domain Shift

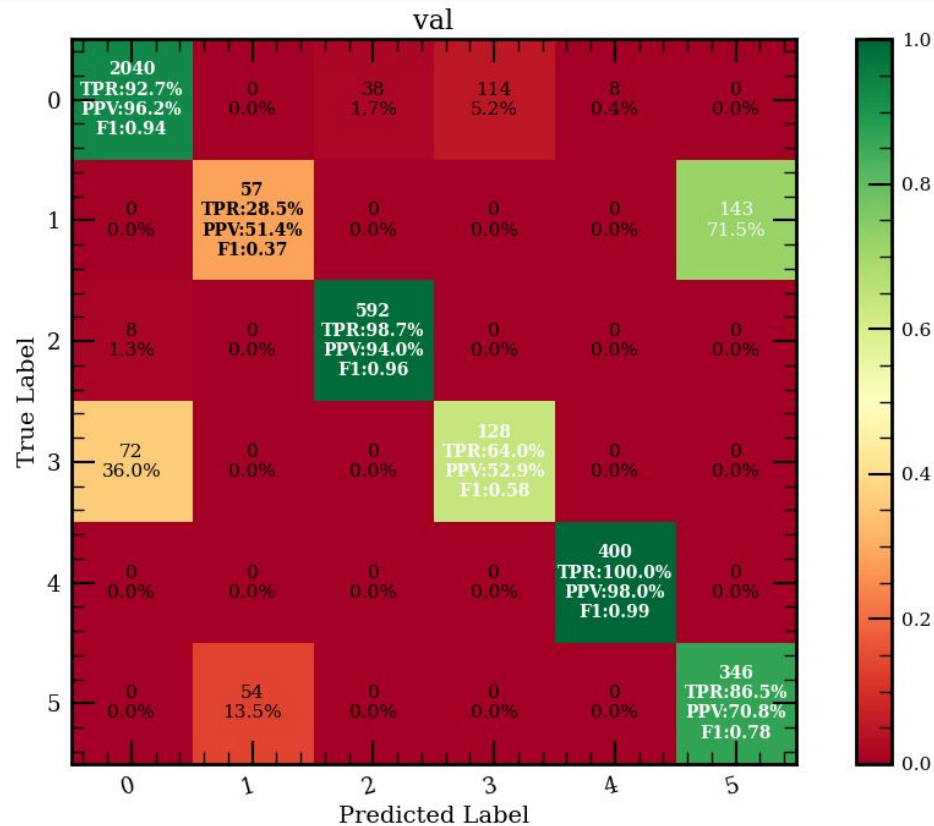
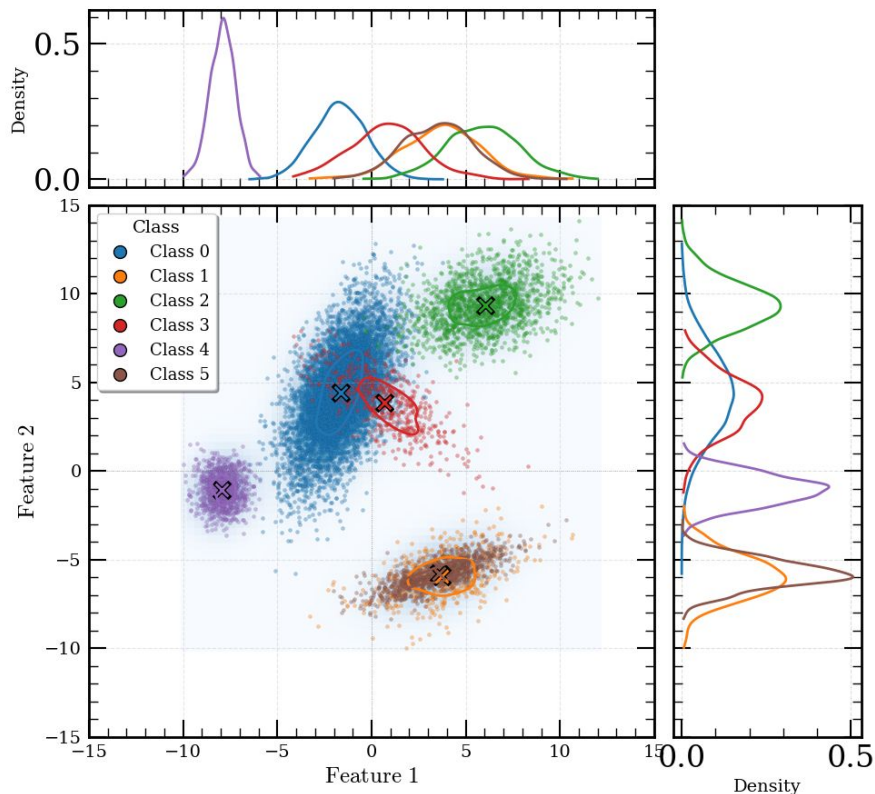


$$H_p(q) = -\frac{1}{N} \sum_{i=1}^N y_i \cdot \log(p(y_i)) + (1 - y_i) \cdot \log(1 - p(y_i))$$

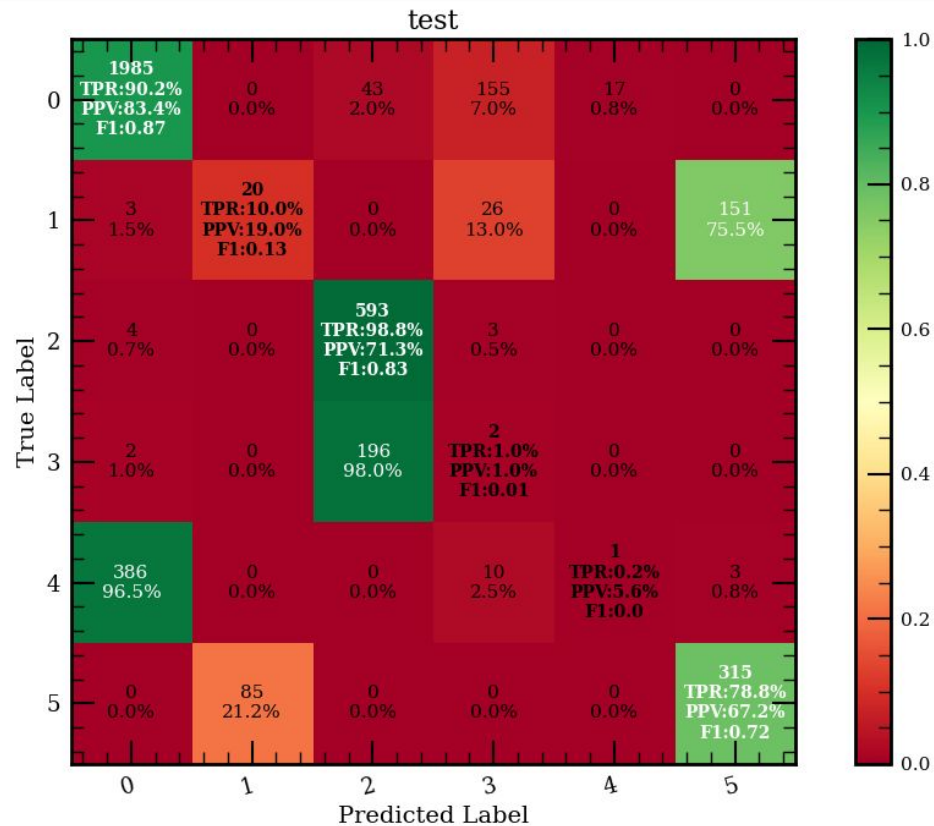
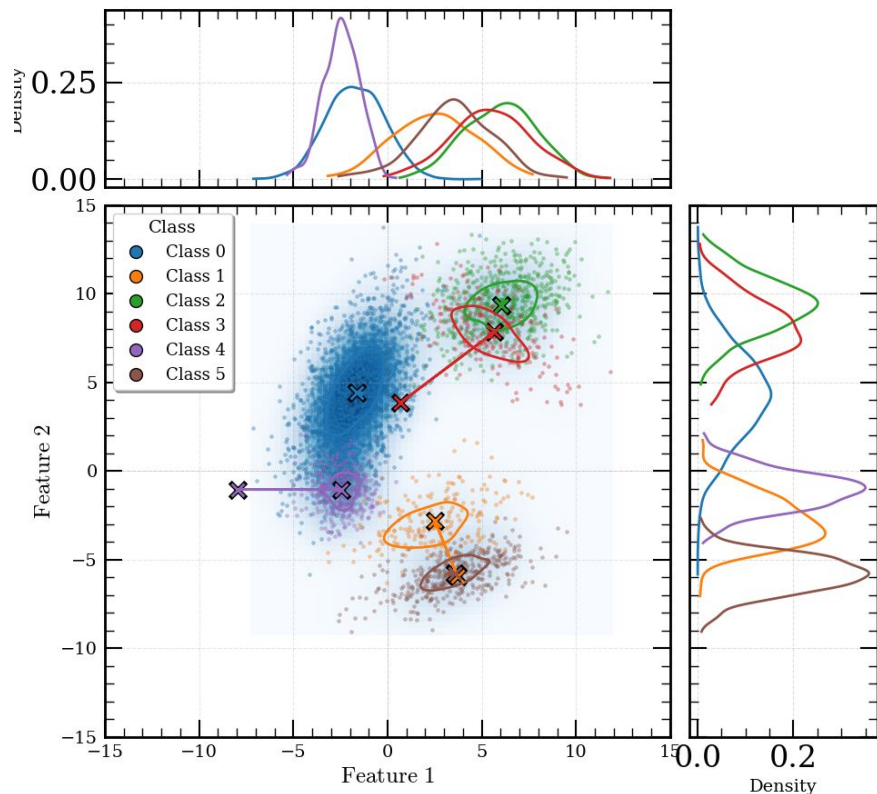
1. Problem: Domain Shift



1. Problem: Domain Shift



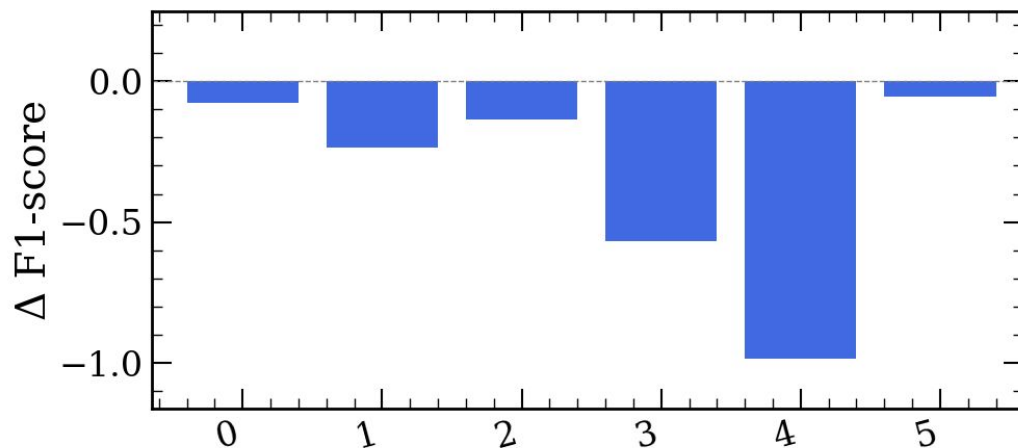
1. Problem: Domain Shift



1. Problem: Domain Shift

=== val vs test Metrics ===

Metric	val	test	Δ (test - val)
Accuracy	0.8908	0.7290	-0.1618
Macro F1	0.7702	0.4277	-0.3425
Macro TPR	0.7840	0.4651	-0.3189
Macro Precision	0.7721	0.4124	-0.3596
Macro AUROC	0.9635	0.7750	-0.1885
Expected Calibration Error	0.0960	0.2278	0.1317
Brier Score	0.2011	0.4950	0.2938



1. Problem: Domain Shift

Training Dataset



Test Dataset



Source Domain

Target Domain



This is "12"

✓ Correct Prediction

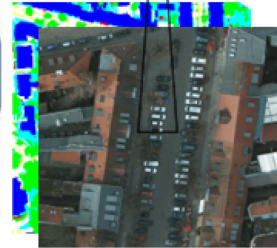


This is "8"

✗ wrong prediction



- Resolution = 5 cm per pixel
- Image sensor: RGB (Red - Green - Blue)
- With pixel-wise labels
- Location: Potsdam



Source domain



- Resolution = 9 cm per pixel
- Image sensor: IRRG (Near Infrared - Red - Green)
- Without pixel-wise labels
- Location: Vaihingen



Target domain

Cross-domain semantic segmentation

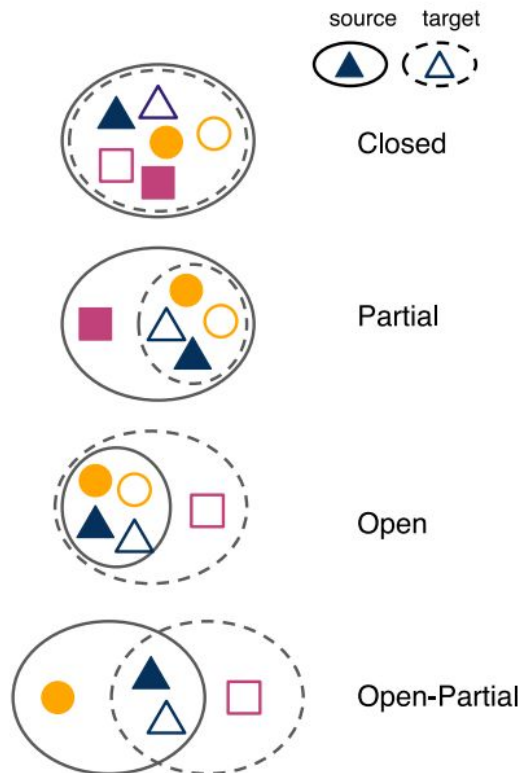


<https://www.mdpi.com/2072-4292/11/11/1369>

2. Solution

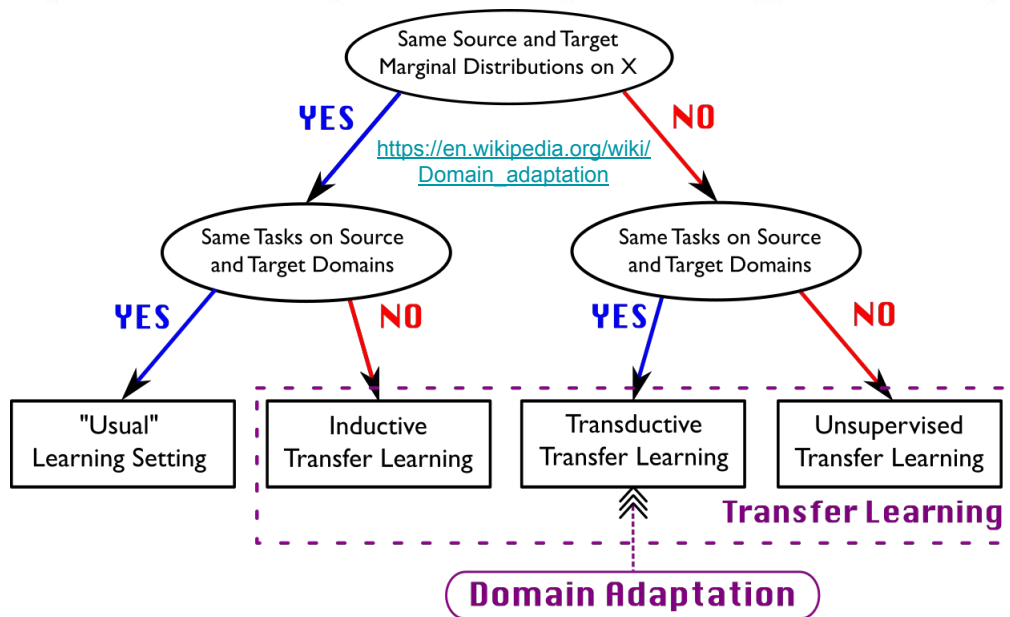
Techniques for dealing with domain shift:
Transfer Learning,
Domain Adaptation,
Contrastive/Self-supervised Learning

2. Solution



<https://arxiv.org/pdf/2302.02005>

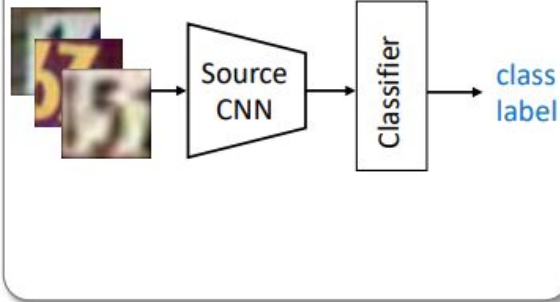
Method	Source Labels	Target Data	Target Labels
Supervised Transfer	✓	✓	✓
Semi-Supervised DA	✓	✓	Few
UDA	✓	✓	✗
DG	✓	✗	✗
CL	✗	Maybe	✗



2. Solution: Domain Adaptation

Pre-training

source images
+ labels

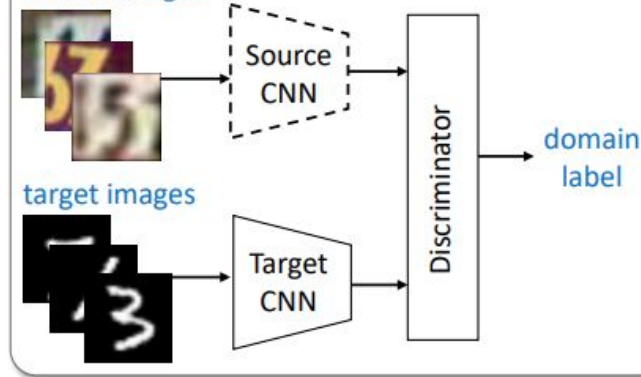


Adversarial Adaptation

source images



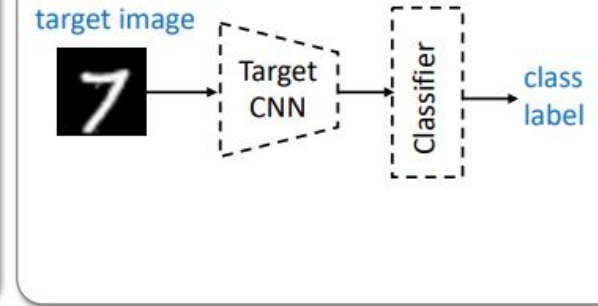
target images



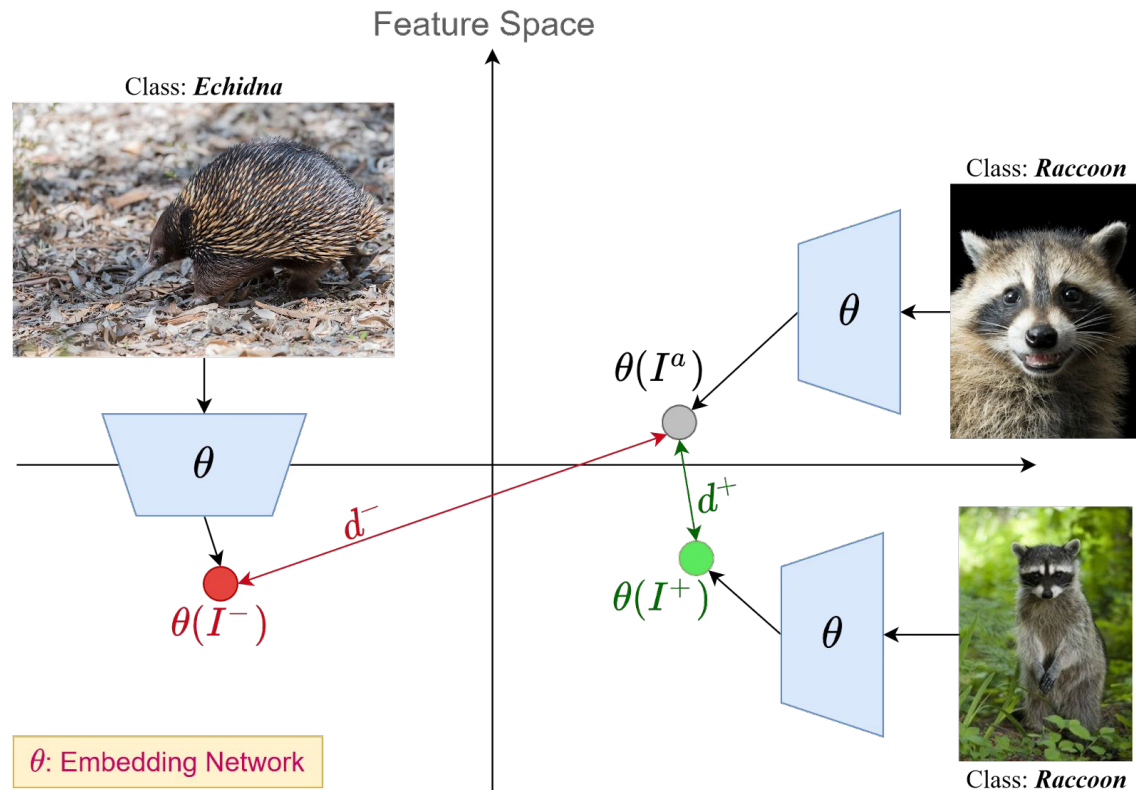
<https://arxiv.org/pdf/1702.05464>

Testing

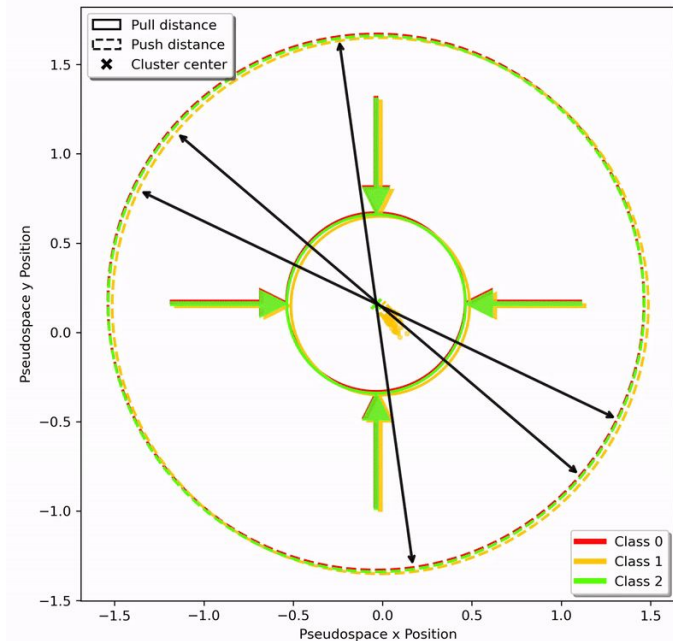
target image



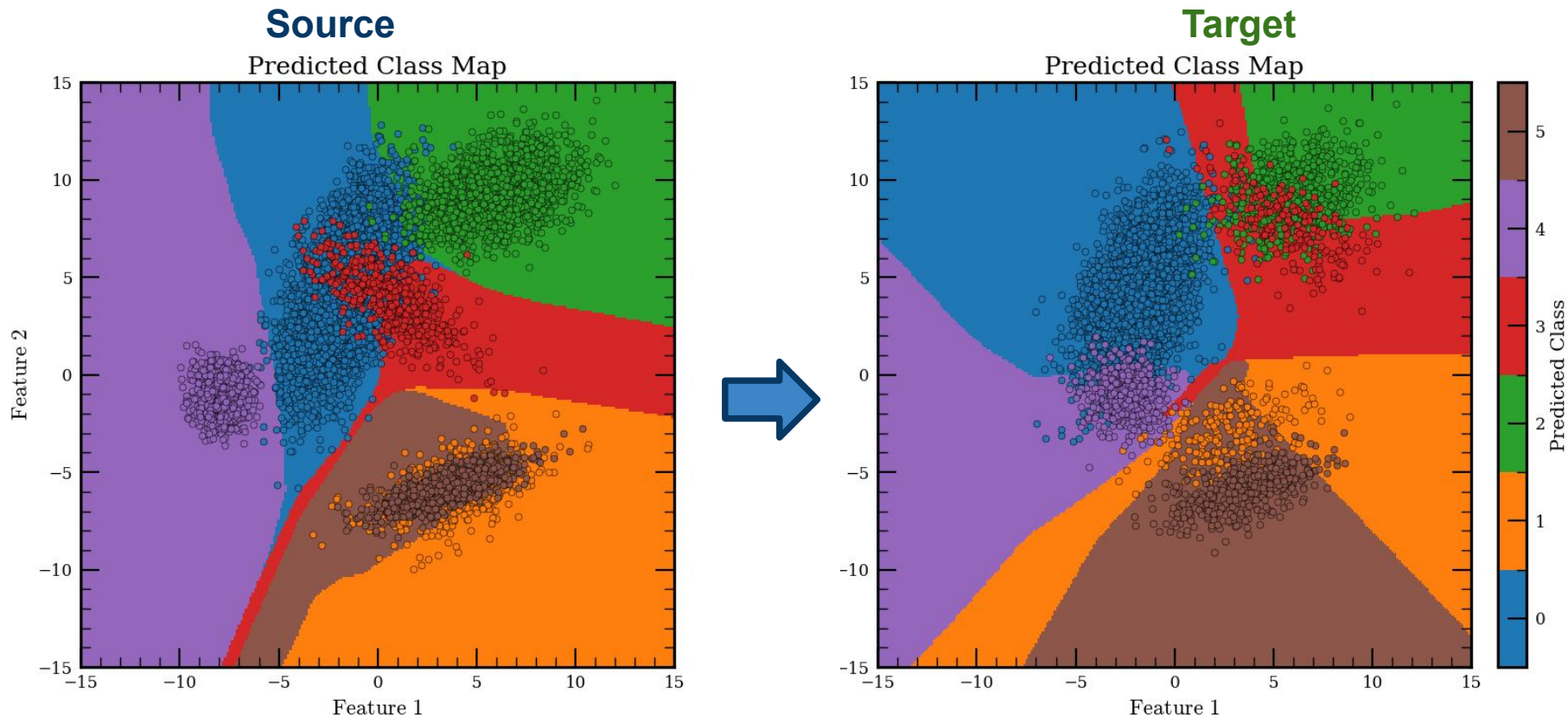
2. Solution: Contrastive/Self-supervised Learning



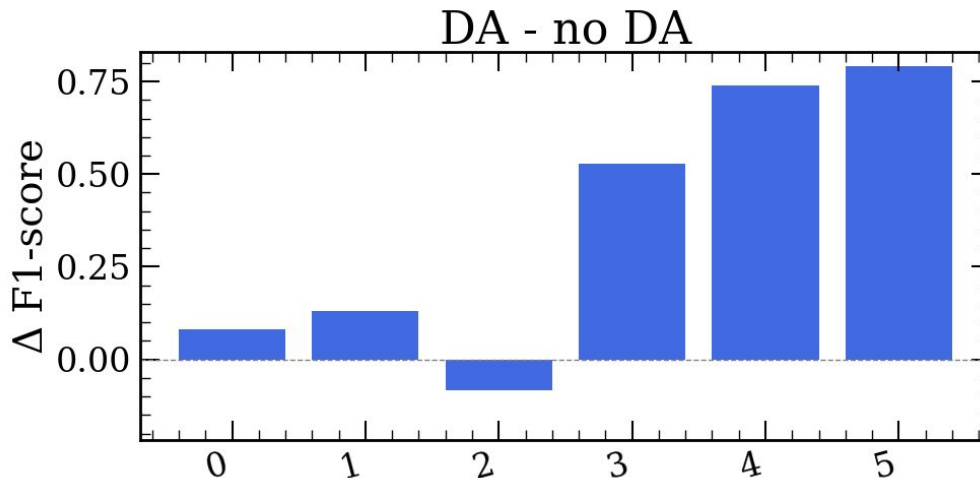
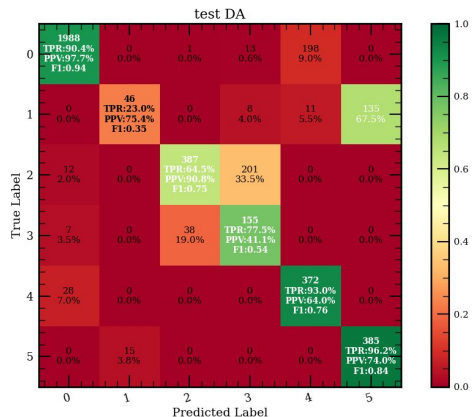
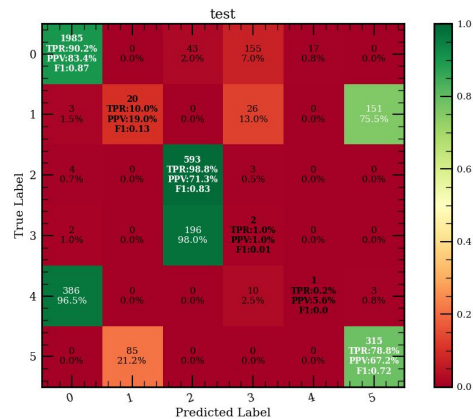
<https://www.v7labs.com/blog/contrastive-learning-guide>



2. Solution: Transfer Learning



2. Solution: Transfer Learning



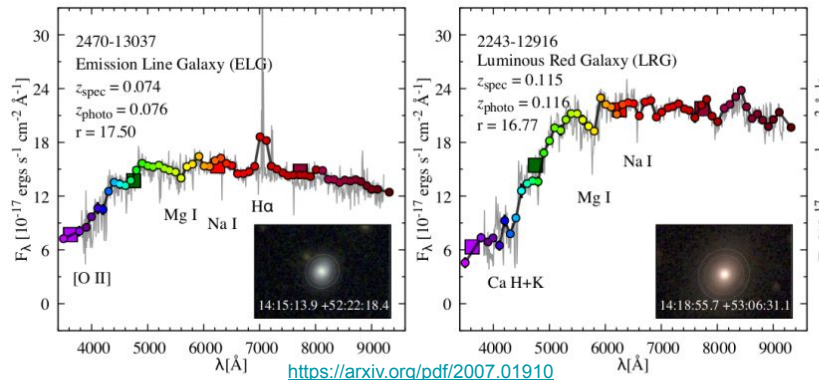
=== no DA vs DA Metrics ===

Metric	no DA	DA	Δ (DA - no DA)
Accuracy	0.6597	0.8333	0.1735
Macro F1	0.3314	0.6964	0.3650
Macro TPR	0.3817	0.7410	0.3594
Macro Precision	0.3426	0.7385	0.3960
Macro AUROC	0.7636	0.9663	0.2027
Expected Calibration Error	0.2722	0.0578	-0.2144
Brier Score	0.5341	0.2576	-0.2765

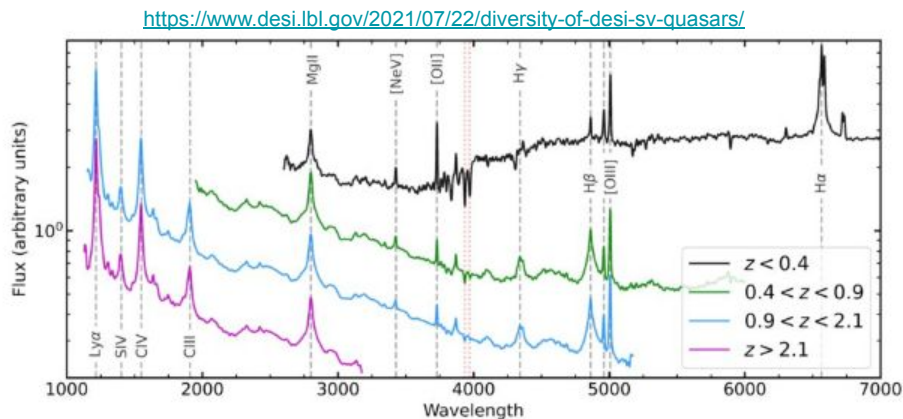
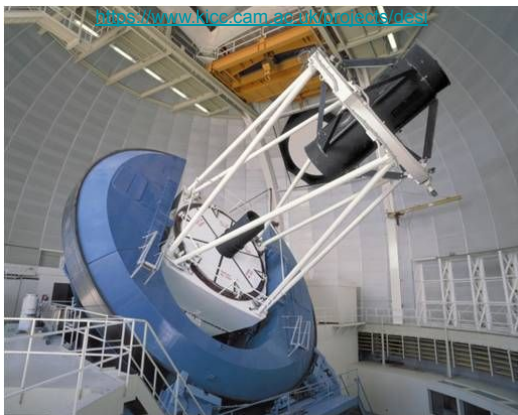
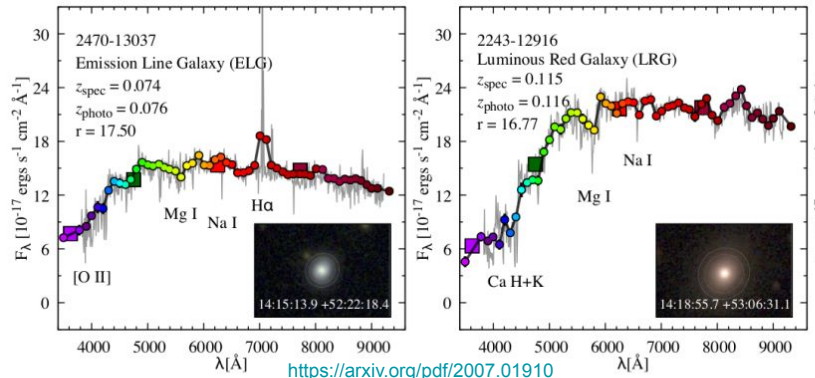
3. Application: From DESI mocks to J-PAS observations

Dealing with our domain shift problem in the context of cosmological surveys

3.1 Data & Experiments



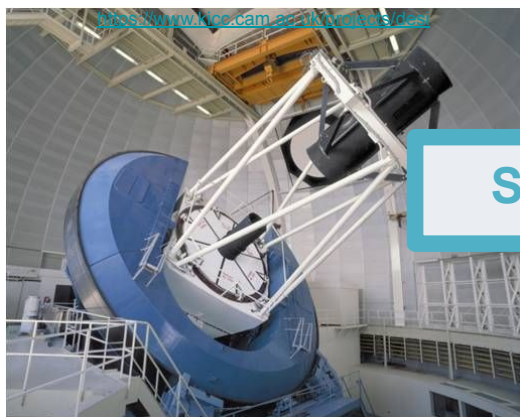
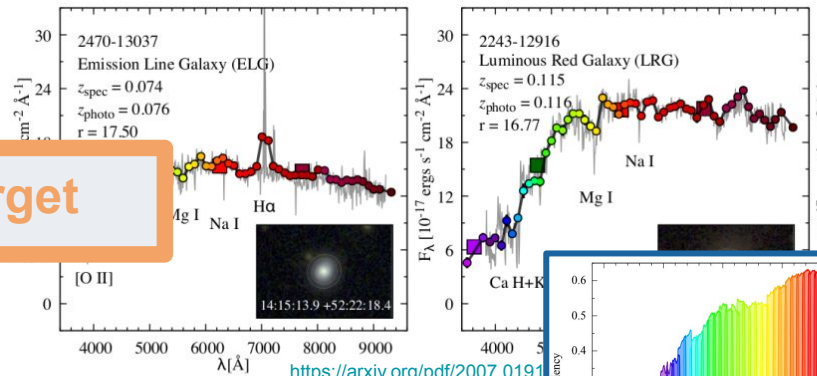
3.1 Data & Experiments



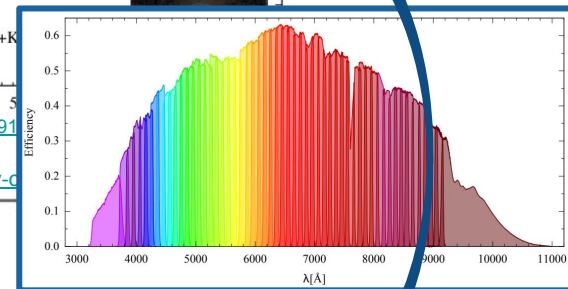
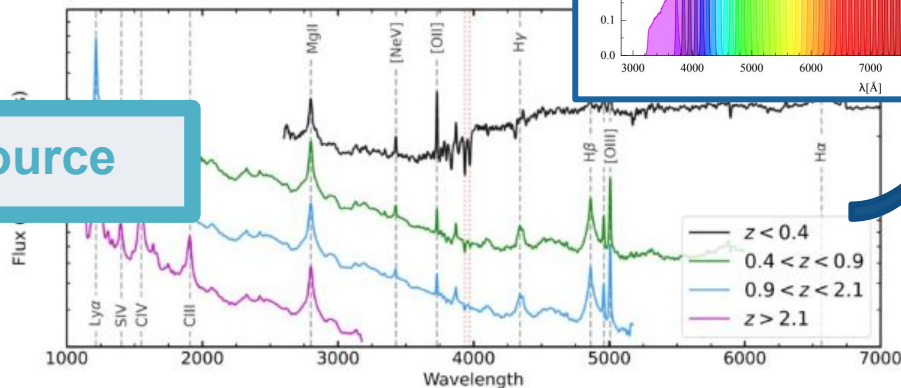
3.1 Data & Experiments



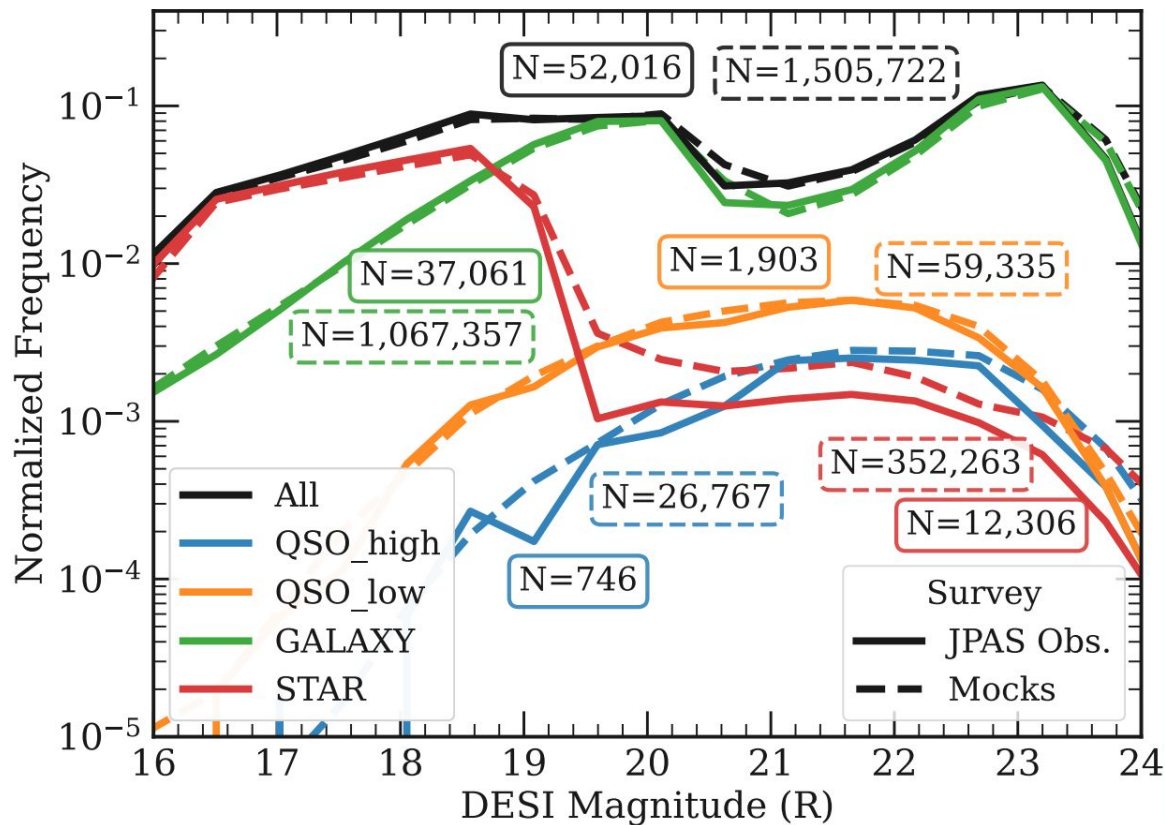
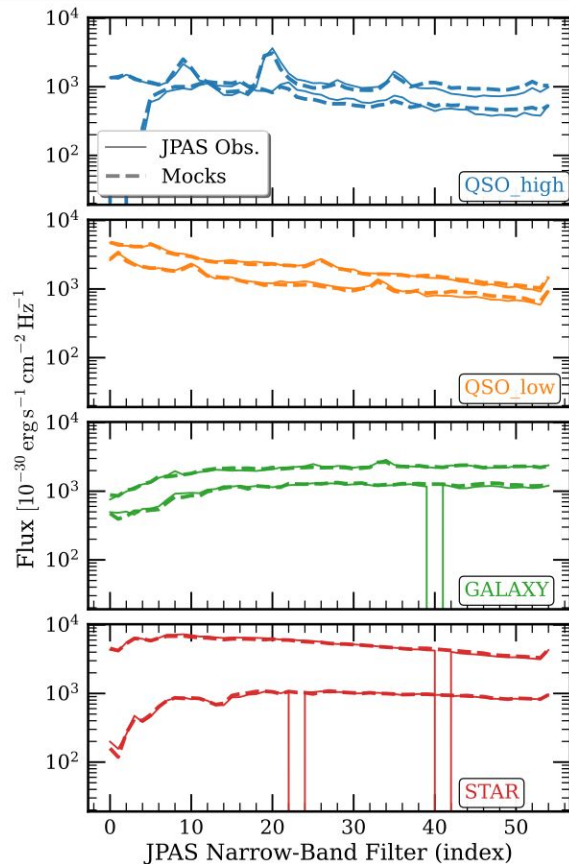
Target



Source

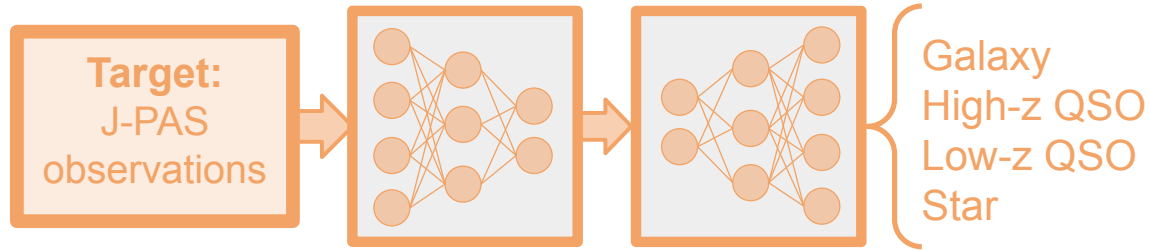


3.1 Data & Experiments



3.2 Method

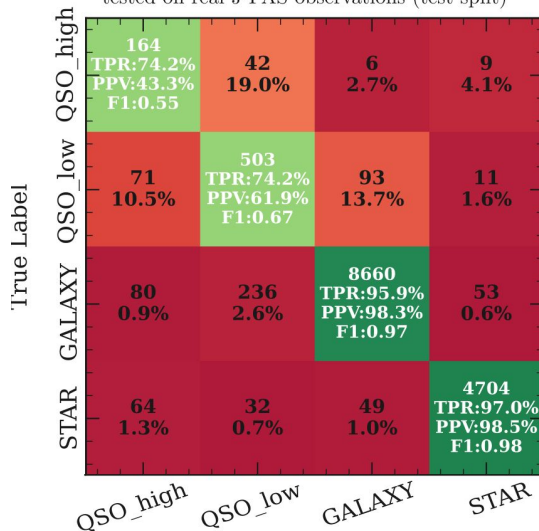
Supervised
model



Supervised model:

trained only with J-PAS observations (training split)

tested on real J-PAS observations (test split)

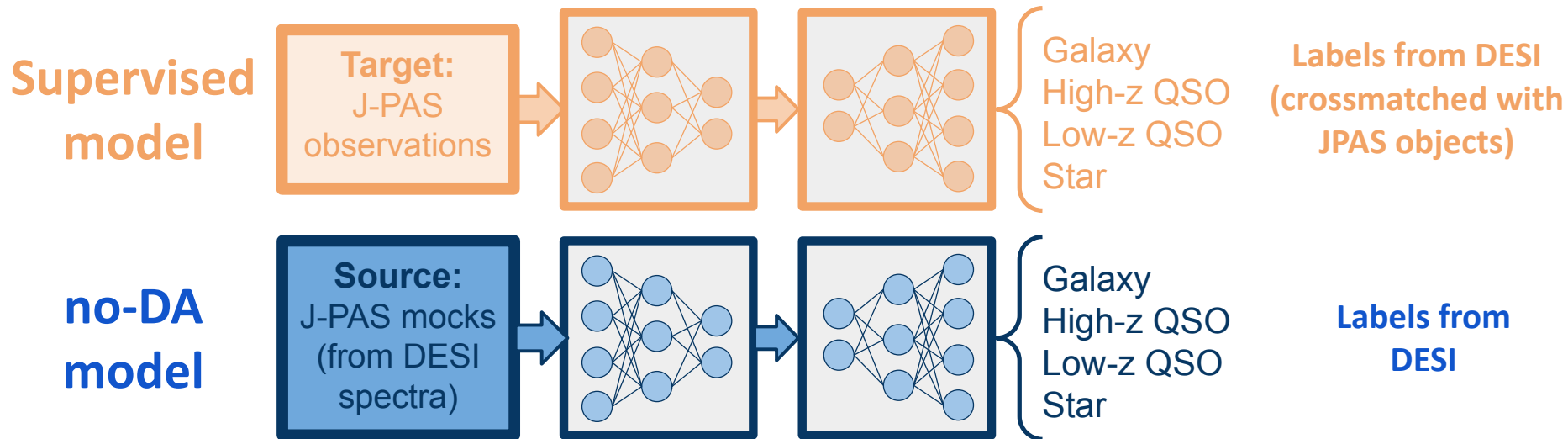


Training ~ 10^4 Sources

Test ~ 10^4 Sources

Labels from DESI
(crossmatched with
J-PAS objects)

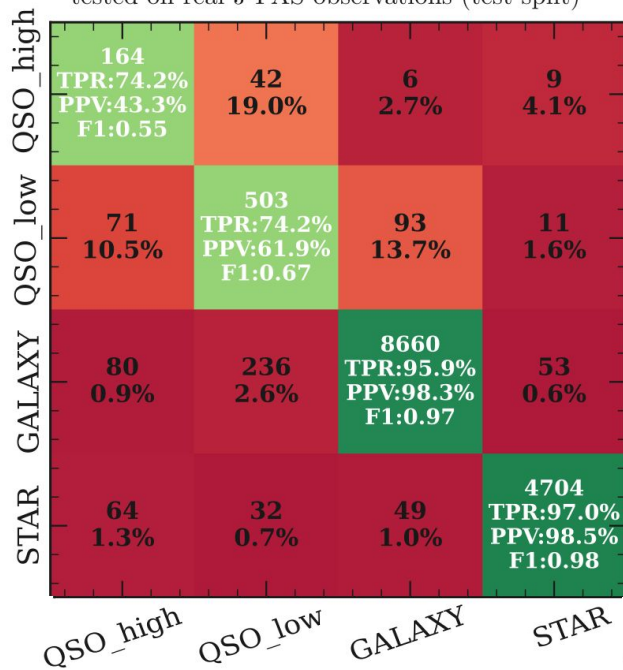
3.2 Method



3.3 Results

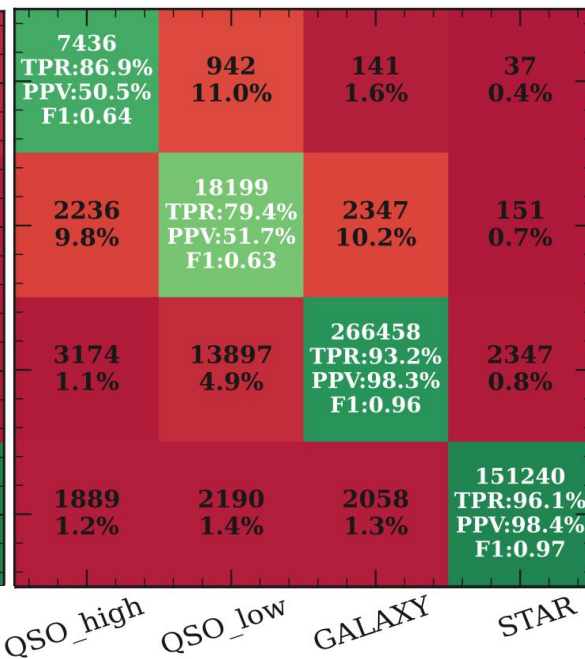
Supervised model:

trained only with J-PAS observations (training split)
tested on real J-PAS observations (test split)



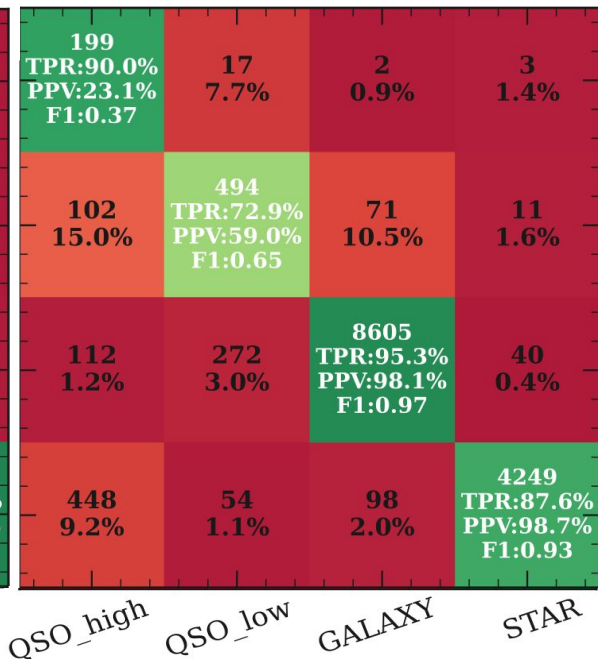
no-DA model:

trained with DESI→J-PAS mocks (training split)
tested on DESI→J-PAS mocks (test split)

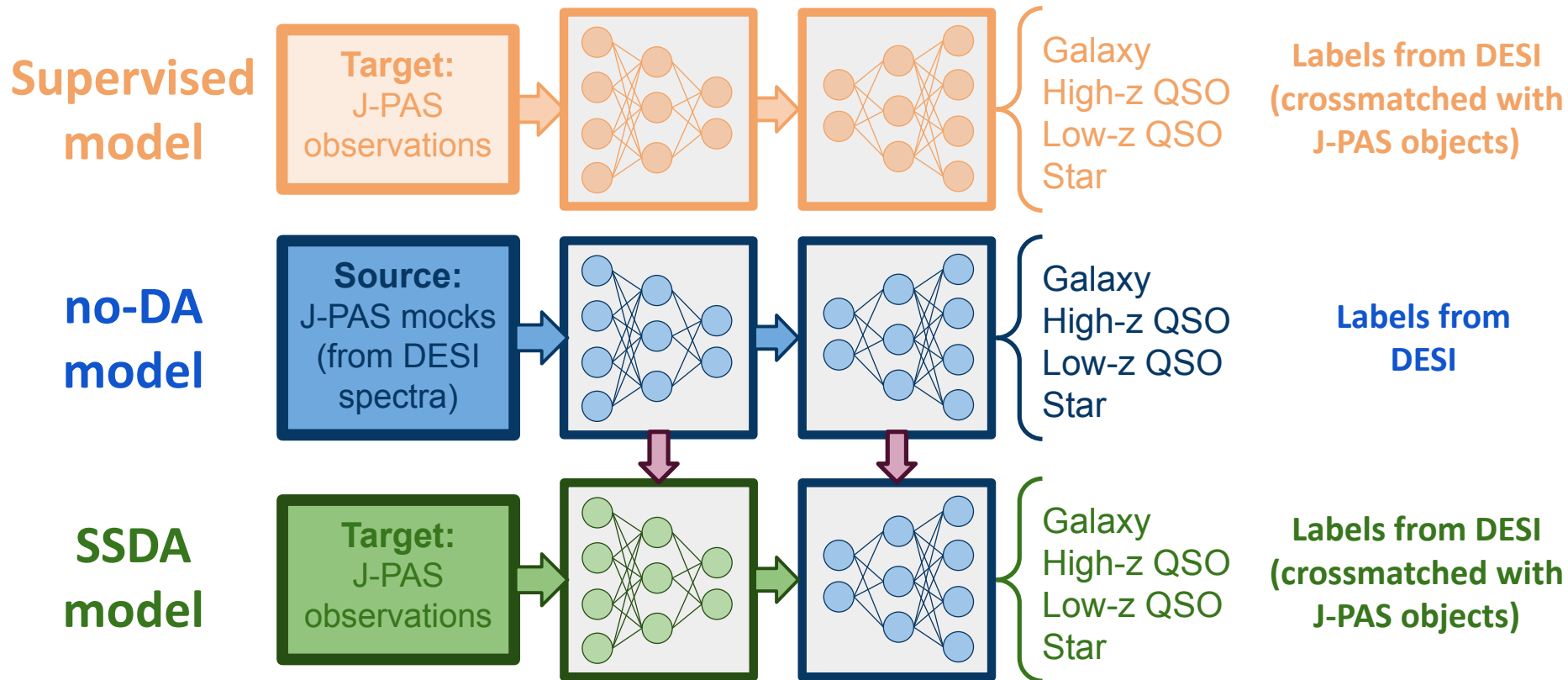


no-DA model:

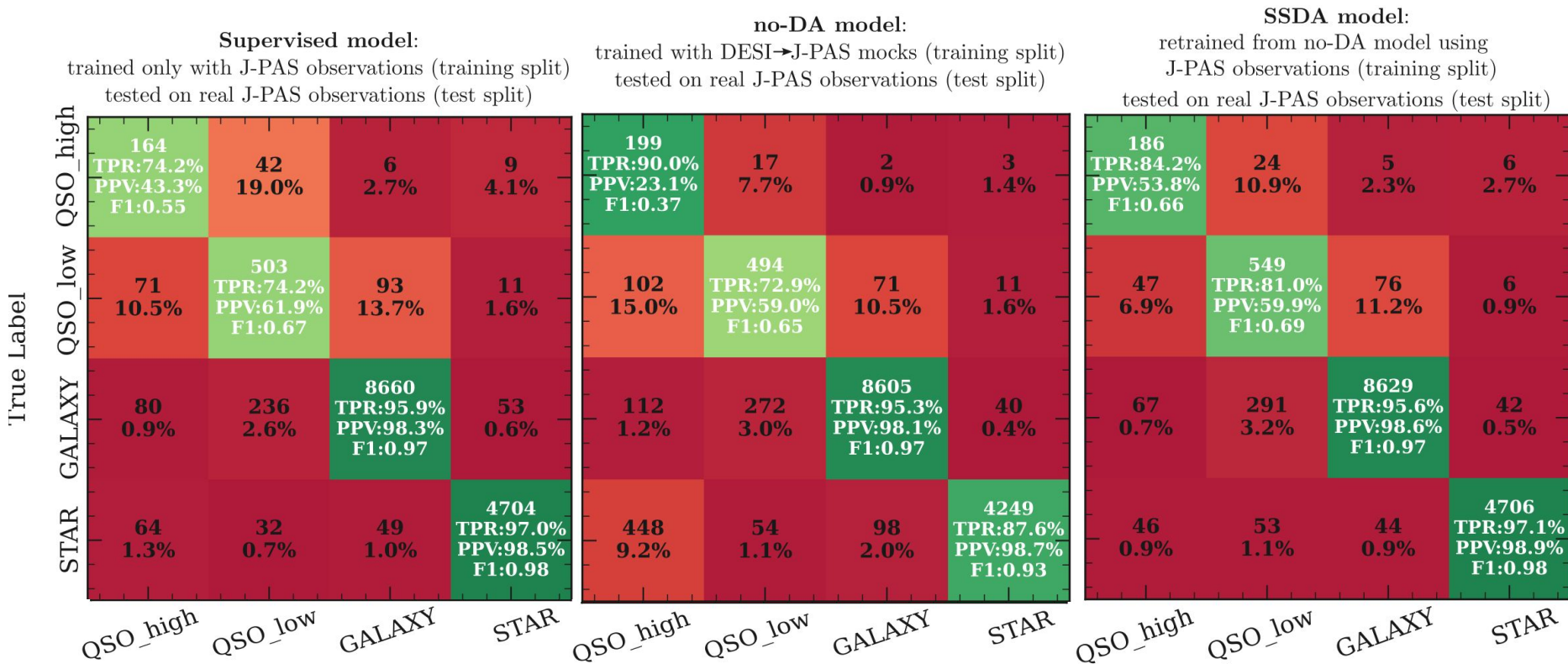
trained with DESI→J-PAS mocks (training split)
tested on real J-PAS observations (test split)



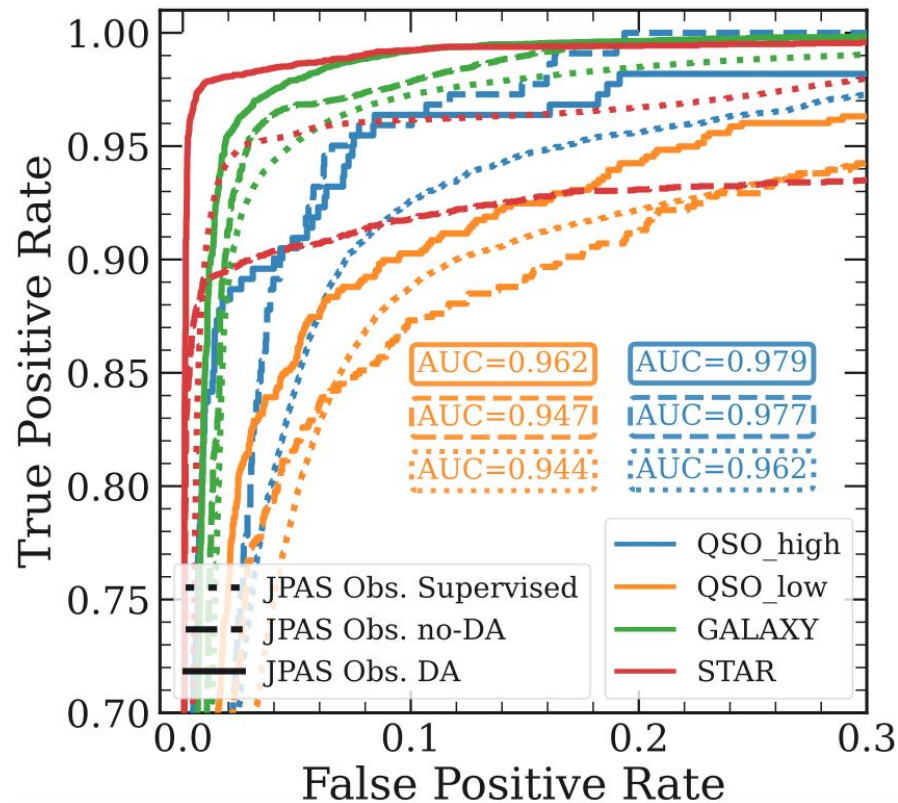
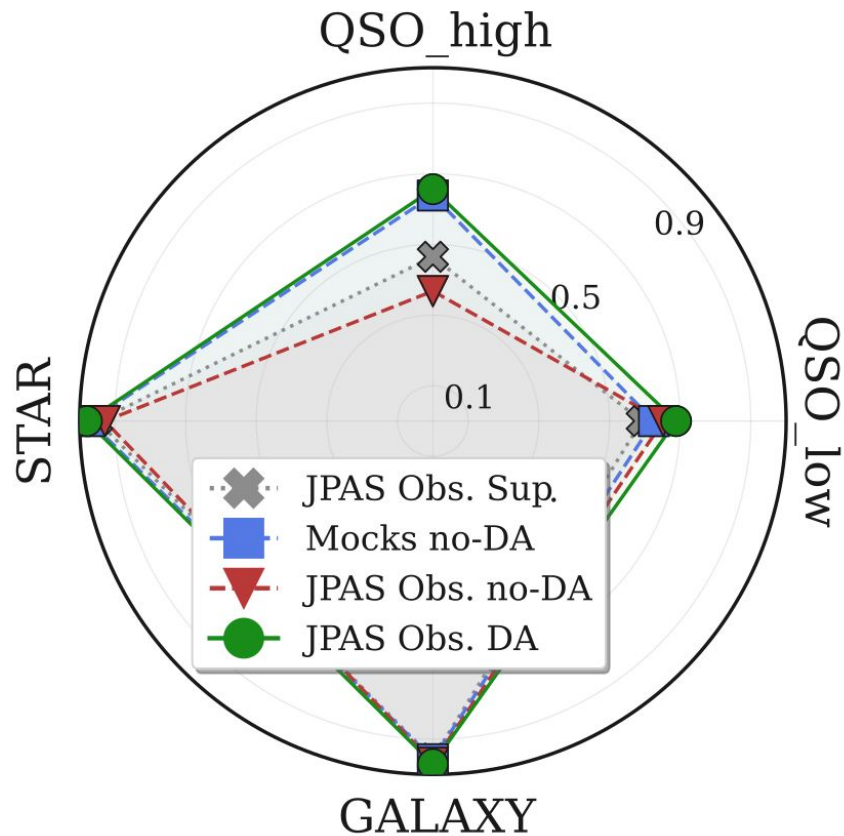
3.2 Method



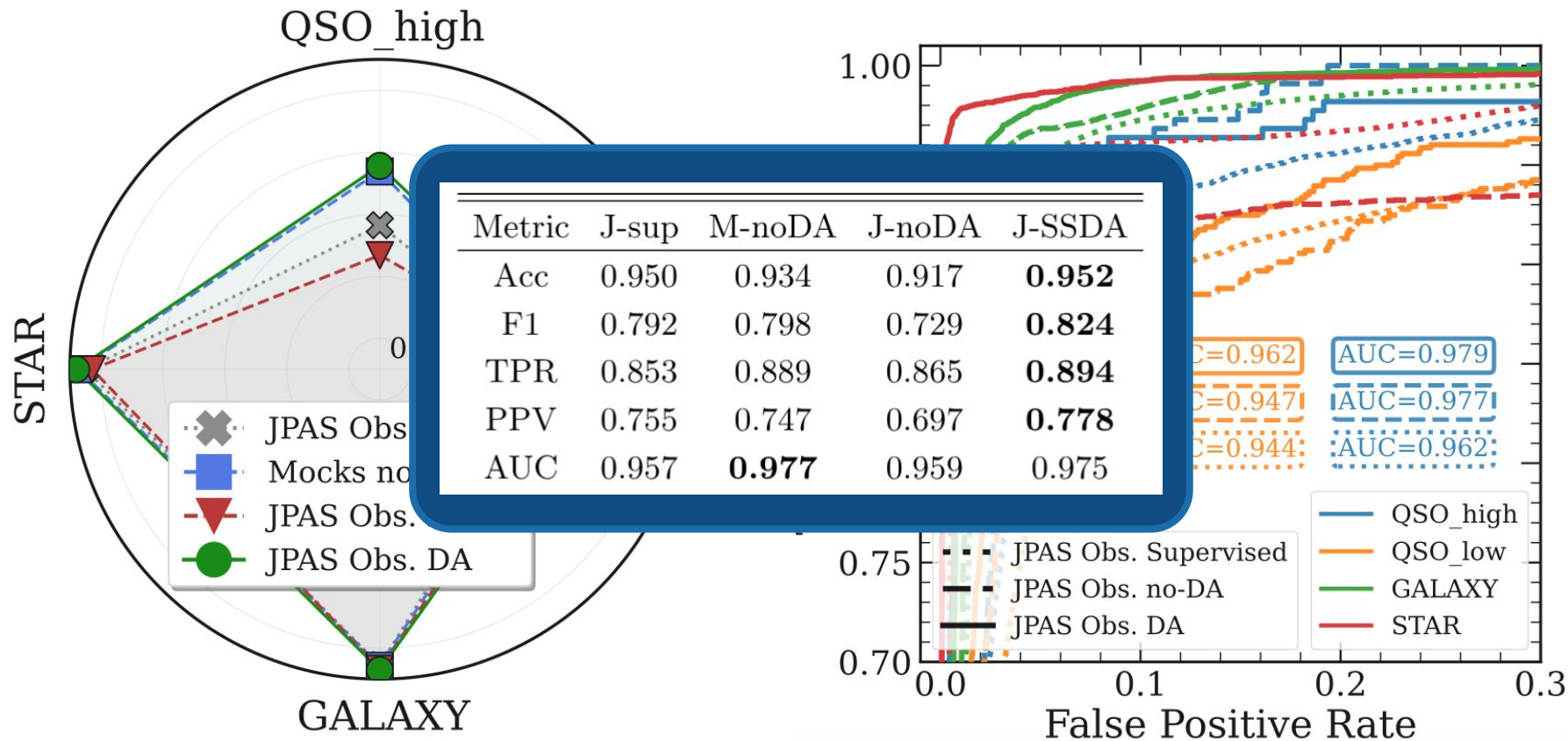
3.3 Results



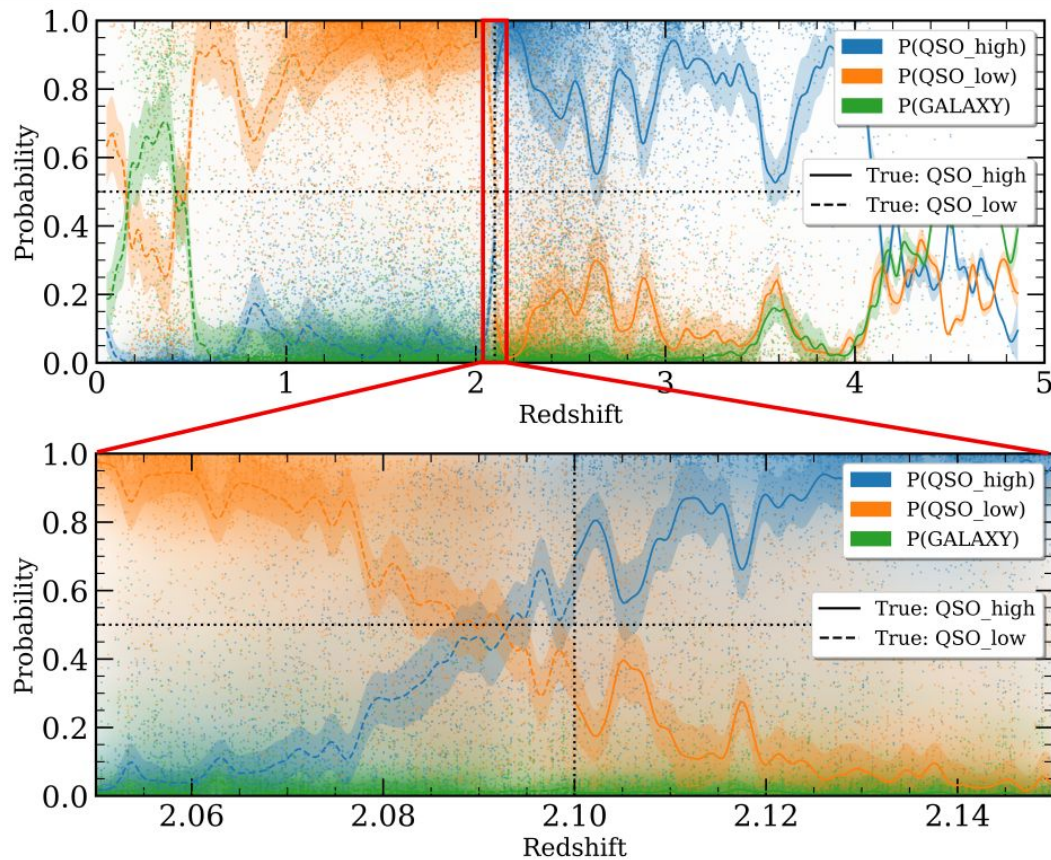
3.3 Results



3.3 Results



3.3 Results



4. Key Takeaways

How can this be useful for your own research?

4. Key Takeaways

Big picture:

Transfer learning, domain adaptation, and contrastive learning are practical ways to fight domain shift problems (both for simulation→observation and cross-survey) that generalize to many areas of physics.

Our case (what we built & got):

- **Public code:** repo now open (https://github.com/daniellopezcano/JPAS_Domain_Adaptation) with reproducible configs and ready-to-run examples.
- **Overall gains:** better target-domain metrics with far fewer labels, reduced bias, and faster cross-survey portability.
- **High-z quasars:** improved purity/recall → cleaner, larger candidate lists for future surveys (e.g., WEAVE).



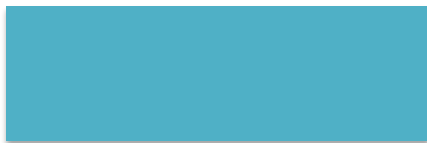
BG #0B1F35



Accent #0F4C81



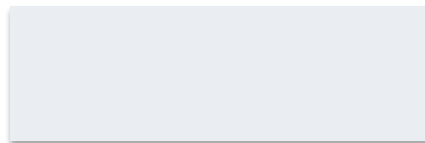
Accent2 #1B6CA8



Accent3 #4FB0C6



Accent4 #F2A365



Ink #EAE EF3